

Generalization of a Ratio Limit for Integral Functions

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August 6, 2026

1 Introduction and Specific Evaluation

We consider the limit problem:

$$L = \lim_{n \rightarrow \infty} \frac{\operatorname{erf}(1/n)}{\operatorname{Si}(1/n)} \quad (1)$$

Recall the integral definitions of the Error Function and the Sine Integral:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad (2)$$

$$\operatorname{Si}(x) = \int_0^x \frac{\sin t}{t} dt \quad (3)$$

Letting $x = 1/n$, as $n \rightarrow \infty$ we have $x \rightarrow 0$. Both numerator and denominator approach zero. Applying L'Hôpital's Rule and the Fundamental Theorem of Calculus:

$$L = \lim_{x \rightarrow 0} \frac{\frac{2}{\sqrt{\pi}} e^{-x^2}}{\frac{\sin x}{x}} = \frac{\frac{2}{\sqrt{\pi}}(1)}{1} = \frac{2}{\sqrt{\pi}} \quad (4)$$

2 Rigorous Generalization

This behavior can be generalized to any ratio of definite integrals whose upper limits converge to zero.

Theorem 1. *Let $f, g : [0, \alpha) \rightarrow \mathbb{R}$ be functions continuous at $t = 0$, with $g(0) \neq 0$. Define the integral functions $F(x) = \int_0^x f(t) dt$ and $G(x) = \int_0^x g(t) dt$. For any sequence (a_n) such that $\lim_{n \rightarrow \infty} a_n = 0$ and $a_n \in (0, \alpha)$, the following holds:*

$$\lim_{n \rightarrow \infty} \frac{F(a_n)}{G(a_n)} = \frac{f(0)}{g(0)} \quad (5)$$

Proof. Let $x \rightarrow 0$. By definition, $F(0) = G(0) = 0$. Since f and g are continuous at 0, F and G are differentiable at $x = 0$ by the Fundamental Theorem of Calculus, with $F'(0) = f(0)$ and $G'(0) = g(0)$.

Using the limit definition of the derivative at zero:

$$\lim_{x \rightarrow 0} \frac{F(x)}{G(x)} = \lim_{x \rightarrow 0} \frac{\frac{F(x)-F(0)}{x-0}}{\frac{G(x)-G(0)}{x-0}} = \frac{F'(0)}{G'(0)} = \frac{f(0)}{g(0)} \quad (6)$$

Since $g(0) \neq 0$, the limit is well-defined. By sequential characterization of limits, substituting $x = a_n$ as $n \rightarrow \infty$ yields the result. $\square$

3 Higher-Order Asymptotic Expansions

We now extend the analysis to cases where the integrands vanish at the lower limit; that is, $f(0) = 0$ and $g(0) = 0$. In such scenarios, Theorem 1 is insufficient, and one must examine the higher-order derivatives or, equivalently, the asymptotic expansions of f and g near $t = 0$.

Assume f and g are smooth in a neighborhood of 0, admitting the Taylor expansions:

$$f(t) \sim c_k t^k + O(t^{k+1}) \quad (7)$$

$$g(t) \sim d_m t^m + O(t^{m+1}) \quad (8)$$

where k and m are the orders of the first non-vanishing derivatives at 0, meaning $c_k = \frac{f^{(k)}(0)}{k!} \neq 0$ and $d_m = \frac{g^{(m)}(0)}{m!} \neq 0$.

Integrating these asymptotic expansions from 0 to x , we obtain:

$$F(x) = \int_0^x (c_k t^k + O(t^{k+1})) dt = \frac{c_k}{k+1} x^{k+1} + O(x^{k+2}) \quad (9)$$

$$G(x) = \int_0^x (d_m t^m + O(t^{m+1})) dt = \frac{d_m}{m+1} x^{m+1} + O(x^{m+2}) \quad (10)$$

The limit of the ratio as $x \rightarrow 0$ depends intimately on the relation between k and m :

$$\lim_{x \rightarrow 0} \frac{F(x)}{G(x)} = \lim_{x \rightarrow 0} \frac{\frac{c_k}{k+1} x^{k+1}}{\frac{d_m}{m+1} x^{m+1}} = \frac{c_k(m+1)}{d_m(k+1)} \lim_{x \rightarrow 0} x^{k-m} \quad (11)$$

This yields three distinct cases:

- **Case 1** ($k > m$): The numerator decays faster than the denominator. The limit is 0.
- **Case 2** ($k < m$): The denominator decays faster than the numerator. The limit diverges to $\pm\infty$, depending on the signs of c_k and d_m and the parity of $m - k$.
- **Case 3** ($k = m$): Both integrands vanish at the same rate. The limit converges to a non-zero finite value:

$$\lim_{x \rightarrow 0} \frac{F(x)}{G(x)} = \frac{c_k}{d_k} = \frac{f^{(k)}(0)}{g^{(k)}(0)} \quad (12)$$

Corollary 1. *If f and g are k -times continuously differentiable at 0, with $f^{(i)}(0) = g^{(i)}(0) = 0$ for all $0 \leq i < k$, and $g^{(k)}(0) \neq 0$, then for any null sequence (a_n) in the domain,*

$$\lim_{n \rightarrow \infty} \frac{\int_0^{a_n} f(t) dt}{\int_0^{a_n} g(t) dt} = \frac{f^{(k)}(0)}{g^{(k)}(0)} \quad (13)$$

This is equivalent to applying L'Hôpital's rule $k + 1$ times.